

Question 1

In bulk RNA-sequencing (Bulk RNA-seq), the expression levels of housekeeping genes generally exhibit a stable positive correlation trend. Based on the provided expression matrix consisting of 12 liver tissue samples and 1,000 genes, we aim to validate this hypothesis using two core housekeeping genes: GAPDH and ACTB.

1. Read the "homework5-Q1.tsv" file and extract the 12 sets of expression values for GAPDH and ACTB; Calculate the Pearson correlation coefficient and Spearman rank correlation coefficient step-by-step using the R programming language (without directly using the `cor()` function).
2. Combining the statistical principles of the two correlation coefficients and the characteristics of the experimental data in this study, explain why there are significant differences between the calculated results of the Pearson and Spearman correlation coefficients.

Question 2

A clinical study measured age (years) and systolic blood pressure (SBP, mmHg) in 25 healthy adults. Use the following R code to create the dataset:

```
age <- c(30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 32, 37, 42, 47, 52,
57, 62, 67, 72, 77, 33, 43, 53, 63, 73)
sbp <- c(115, 122, 128, 132, 140, 148, 155, 160, 168, 175, 118, 124,
130, 135, 142, 150, 157, 162, 170, 178, 120, 131, 140, 151, 161)
bp <- data.frame(age, sbp)
```

- a. Fit a simple linear regression model with sbp as the response and age as the predictor.
- b. Test the significance of the regression coefficient for age at the 5% level. State the null and alternative hypotheses, the test statistic, the p-value, and your conclusion in context.
- c. Compute and interpret a 95% confidence interval for the slope coefficient.

Question 3

Using the built-in R dataset 'mtcars', you are required to build a multiple linear regression model. The target is to explore how a vehicle's structural characteristics affect its fuel efficiency.

1. Write the R code to construct the multiple linear regression model described above.
2. Write R commands to extract and print the individual p-values for the regression coefficients of 'wt' and 'hp' (from the t-tests).

3. Write R commands to dynamically compute and print the overall model significance p-value (from the F-test).
4. Based on the statistical outputs, analyze the results: Compare the overall model significance with the individual coefficient significances. Explain any potential statistical paradox or phenomenon observed (e.g., why a variable might become insignificant in a joint model).

Question 4

Facemire et al. examined changes in the ratio of blood-level estradiol to blood level testosterone (each in pg/ml) in male panthers with age(in years). The data are as follows:

```
Age <- c(1,2,3,4,5,6,7,8)
```

```
Hormone_Ratio <- c(0.42, 0.39, 0.53, 0.65, 0.75, 0.56, 0.53, 0.48)
```

1. Generate a scatter plot with “ln(hormone ratio)” on the y-axis and age on the x-axis. Does the relationship between the two variables appear to be linear or non-linear?
2. Fit the following straight-line regression model: “log(hormone ratio)~age” Plot the line on your scatter plot. Does your linear model in age significantly predict hormone ratios at $\alpha = 0.05$?
3. Add “age^2” to the model in part 2. Plot the least squares quadratic curve on your scatter plot. Does adding “age^2” to your model significantly improve prediction of hormone levels? Hint: Use ANOVA

Question 5

In this question, we will investigate the risk factors associated with low infant birth weight. We will use the birthwt dataset from the MASS package. The outcome of interest is “low” (an indicator of whether the birth weight was less than 2.5 kg).

Please load the dataset using “library(MASS); data(birthwt)”. Note that some categorical variables in this dataset are coded as integers. You must convert variables “race”, “smoke”, and “ht” to factors before fitting the model.

- a. Fit a multiple logistic regression model predicting “low” using “age”, “lwt” (mother’s weight at last menstrual period), “race”, “smoke”, and “ht” as predictors. Display the summary of the model.
- b. Then try to reduce the number of features from the model in (a) by keeping all the significant features and make another regression model. (Use 0.05 as significant level)
- c. Use proper test to test the difference between these two models. (Use 0.05 as

significant level). Hint: use "anova()"

- d. You can randomly sample 100 rows as training data to make a regression model using all variables mentioned in (a), and then take remaining 89 rows as prediction data to calculate the accuracy of your model. The predict function return a value between 0 and 1, < 0.5 belong to the normal weight, ≥ 0.5 belong to the low weight. (Use "set.seed(320)" to explicitly define random seed)